Electricity demand has not become more price-responsive despite ninety years of technological change*

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Abstract

Energy planners have long assumed that electricity demand will grow more price-responsive as metering, automation, and storage spread, an assumption now embedded in decarbonization plans. We test it against the empirical record: 4,720 own-price elasticity estimates from 462 studies, with data spanning 1934–2024, ranked on a single ladder of identification quality from naive regressions to randomized experiments. Three findings emerge. First, better-identified studies find *smaller* responses: the publication-bias-corrected short-run elasticity is about -0.16 (a 10% rise in the electricity price cuts consumption by under 2%), and only -0.09 among the best-identified studies, whose adjusted value is statistically indistinguishable from zero. Second, responsiveness grows with time to adjust, roughly doubling from -0.16 in the short run to -0.38 in the long run as the capital stock turns over, but this pattern has itself been stable for decades. Third, and most important, responsiveness shows no upward trend across nine decades of data; if anything, the most technology-rich settings, including time-of-use pricing, are the *least* price-responsive in total consumption. Demand flexibility will have to be engineered and paid for; nothing in the historical record suggests that prices alone will deliver it.

Keywords: electricity demand, price elasticity, meta-analysis, identification, publication bias, demand flexibility

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1 Introduction

Surveying, in 1981, the fifteen residential time-of-use pricing experiments the U.S. Department of Energy had run since 1975, Dennis Aigner recorded the profession's expectation: with a full commitment to time-varying prices, "appliance choices will be made with an eye to TOU response; new appliances will become widely available," and the pricing strategy "surely must be even more desirable in the long-run" (Aigner, 1985, p. 39). Four decades later, a recent working paper synthesizing time-based electricity rates opens with the same expectation, now attached to decarbonization: time-based rates "may be a useful and efficient tool for encouraging demand-side flexibility" in the transition to a high-renewable grid, and how the response will change "with widespread adoption of emerging technologies" remains, in the authors' words, an open prediction problem (Kahn-Lang *et al.*, 2025). Between these two statements, the assumption that price responsiveness of electricity demand is about to rise has underwritten smart-metering business cases (Faruqui & Sergici, 2010), demand-response program design (Torriti, 2014), and the demand-side assumptions of energy-system models (Huntington *et al.*, 2019). It is among the more consequential quantitative assumptions in energy policy: if demand responds strongly to prices, decarbonized power systems need less storage and less firm capacity; if it does not, plans that rely on flexible demand will fall short.

The assumption is testable, and in the four decades since it was first recorded nobody has tested it head-on (Table S3 records it in the words of those who hold it). This paper brings the entire empirical record to bear on it: 4,720 own-price elasticity estimates from 462 studies, with underlying data spanning 1934–2024, to our knowledge the largest dataset assembled on the price elasticity of electricity demand, and one designed so that the question "has responsiveness changed?" can be separated from the two confounds that would otherwise contaminate the answer.

The first confound is identification. Reported elasticities depend on how a study confronts the simultaneity of price and quantity: tariff schedules respond to consumption, regulators respond to demand growth, and average prices constructed from bills divide revenue by the quantity being explained (Alberini & Filippini, 2011; Ito, 2014). A literature whose identification standards improve over time will exhibit spurious elasticity "trends" driven by changing methods rather than changing behavior. We therefore place every estimate on a single identification ladder, running from *design-based* studies at the top (randomized experiments, mandated natural experiments, and difference-in-differences) down through instrumented specifications, panel fixed effects, and naive regressions, in the spirit of the design-quality tiers that Kahn-Lang *et al.* (2025) apply to pricing pilots. The second confound is selective reporting. Statistically significant, correctly signed elasticities are easier to publish, which correlates estimates with their standard errors and inflates naive averages (Stanley *et al.*, 2008; Ioannidis *et al.*, 2017; Brodeur *et al.*, 2020). If selection pressure has itself drifted over time, so has the visible literature, independently of consumers. We therefore run modern selection corrections within tiers and horizons, and we separate two clocks that previous surveys usually conflated: the vintage of a study's *data* (do consumers behave differently?) and the study's *publication* date (does the literature report differently?).

First, *credibility shrinks the elasticity.* Moving up the ladder, the short-run elasticity falls in magnitude from -0.37 (raw naive mean) to -0.09 among design-based estimates (mean beyond bias). An elasticity of -0.37 means a 10% rise in the electricity price cuts consumption by under 4%; at -0.09 the same rise cuts it by under 1%. Both fall well short of the 10% price change, but the gap between them is large: the best-identified estimate is about a quarter the size of the naive one, shrinking a low response to almost none. Once observable study characteristics are held fixed, the design-based tier is statistically indistinguishable from zero ($+0.02$, $[-0.21, 0.25]$) and significantly less elastic than the naive baseline ($p = 0.006$). The instrumented tier, by contrast, is the one identification upgrade that remains as elastic as the naive literature (-0.19 adjusted, against -0.24 naive): instrumenting produces essentially none of the movement toward the experimental benchmark seen elsewhere on the ladder. Selective reporting differs sharply across the tiers: the observational literature shows pronounced funnel asymmetry (FAT $= -0.85$, $p < 0.001$), while the design-based sample shows none ($p = 0.97$). The corrected short-run consensus is small: bias-corrected estimates range from -0.04 to -0.23 across correction philosophies, with our preferred estimate at -0.16.

Second, *responsiveness grows with adjustment time, not with calendar time.* The bias-corrected elasticity roughly doubles from the short run (-0.16) to the long run (-0.38), a gradient consistent with the within-study adjustment path that Deryugina *et al.* (2020) recover from a municipal-aggregation natural experiment, where the elasticity triples from -0.09 after six months to -0.27 after two years, and with the mechanism that Costa & Kahn (2011) identify in the housing stock: the long-run response operates through durable capital, so it arrives with the turnover of buildings and appliances, and it plateaus at the substitution possibilities the stock allows. The gradient itself is stable across eras of data: in every era since the mid-1970s the corrected long-run value has sat between -0.29 and -0.47. The long-run response has already arrived, at about -0.38, well short of -1, the unit-elastic level a fully flexible demand would reach.

Third, *the elasticity shows no trend in the data vintage.* The short-run trend per decade of data is statistically indistinguishable from zero in every specification: the raw record, the bias-corrected record, and the fully adjusted record (both the data and publication clocks plus identification and composition moderators). The no-trend outcome is not just because the data are noisy: in the specifications precise enough to be informative, even the outer edge of the 95% interval reaches only two-thirds to four-fifths of the way to a doubling of responsiveness over three decades, the canonical flexibility scenario. The point estimate in the fully adjusted specification actually leans toward *less* elastic demand ($+0.017$ per decade, $p = 0.53$). Estimates from time-of-use settings, the environments at the heart of the flexibility premise, are closer to zero than the rest of the literature (coefficient $+0.14$, $p = 0.007$), and the design-based and time-of-use cells are consistent with the pilot-experiment literature, the -0.02 to -0.10 range of Faruqui & Sergici (2010) and the -0.075 average of Kahn-Lang *et al.* (2025).

This exercise is a descriptive cross-study audit of a literature, anchored by design-based benchmarks and within-study contrasts where they exist. No individual study spans ninety years or six identification tiers, so only a pooled reading of the whole record can ask whether the corrections the literature trusts reproduce what clean designs deliver, and whether the parameter planners extrapolate has ever moved. Our contribution is therefore a credibility audit of the assumption that the consensus elasticity is about to change, in the tradition of meta-research on selective reporting and the reliability of empirical economics (Ioannidis *et al.*, 2017; Brodeur *et al.*, 2020; DellaVigna & Linos, 2022; Christensen & Miguel, 2018).

Prior syntheses have examined the individual pieces but not the question itself. Espey & Espey (2004) include publication-year and data-vintage dummies in a 36-study sample and find drift; Labandeira *et al.* (2017) report, in a single sentence, that a technical-progress trend "is not significant at any level in any given specification"; Zabaloy & Viego (2022) find significant data-vintage drifts of opposite signs for short- and long-run elasticities in Latin America; Zhu *et al.* (2018) bin studies by period with no correction machinery; the surveys of Dahl (1993, 2011), Fatima (2023), and Marques *et al.* (2024) report static summaries. None makes the time path its central question, none runs it with bias correction, identification controls, and clustered inference, and none but Espey & Espey (2004) separates the data clock from the publication clock, the separation that, on our corpus, can account for the contradictory findings of these syntheses.

The paper proceeds as follows. Section 2 describes the corpus and the three central variables. Section 3 compares estimates across the identification ladder. Section 4 audits selective reporting along it. Section 5 estimates the adjustment-horizon path. Section 6 runs the calendar-time test and states the bound. Section 7 translates the results into the numbers planners and modelers should use. Section 8 concludes.

2 Data

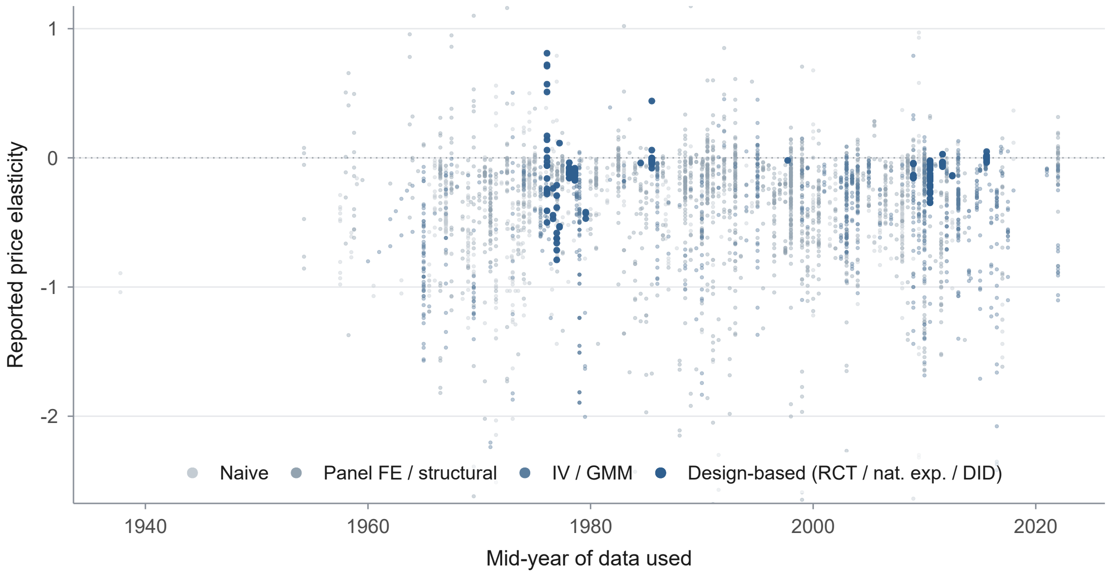
Corpus. We collect estimates of the own-price elasticity of electricity demand together with their standard errors from published studies and working papers. The final dataset contains 4,720 estimates from 462 studies (listed in Supplementary appendix S2), each reporting a usable measure of uncertainty. Our inclusion rule is deliberately simple: an estimate enters only if it reports a standard error, or a t-statistic or p-value from which one can be recovered; an estimate without any such measure cannot be corrected for publication bias and is excluded. A small number of estimates are methodologically borderline (notably: the construction-vintage elasticity of Costa & Kahn, 2011; a separate, author-disavowed wrong-signed estimate; and an aggregate event-study elasticity); they carry a usable standard error and so meet our inclusion rule, and are individually immaterial at this sample size. Publication dates run from 1951 to 2026; the underlying data run from 1934 to 2024 (median data mid-year 1988). Each estimate carries 55 coded characteristics covering the data (country, sector, aggregation, frequency), the demand model (functional form, dynamics, controls), the price variable (average, marginal, time-of-use), the estimation method, and the publication outlet.

Our baseline sample places all comparable own-price elasticities on a common Marshallian (uncompensated) footing: the 3,324 estimates from 366 studies (1,647 short-run, 954 intermediate-run, 723 long-run) that carry a usable standard error and are either reported directly as Marshallian or converted from a compensated (Hicksian) estimate. The conversion applies the Slutsky relation $\varepsilon_M = \varepsilon_H - s\eta$ with electricity budget share $s = 0.04$, a representative value for the electricity share of household spending in consumer expenditure surveys (U.S. Bureau of Labor Statistics, 2024), and each estimate's own income elasticity at the matching horizon. Because s is small, the conversion shifts each estimate by roughly 0.01 and leaves standard errors essentially unchanged (the delta-method adjustment is negligible), so the directly reported and the converted estimates can be read on a single scale.

Restricting to the 2,579 directly reported Marshallian estimates alone (the *pure-Marshallian* sample; Supplementary appendix S4) leaves every headline result unchanged, as does varying s between 0.02 and 0.08. A further 1,396 corpus estimates are held out of the baseline, either not convertible to the Marshallian footing or with an indirectly derived elasticity or standard error (Supplementary appendix S1); Table S5 shows they are statistically indistinguishable from it. Effects and standard errors are winsorized at 1% throughout, and main-text inference clusters at the study level, with the model-averaging and full-battery appendices (Supplementary appendix S6, Supplementary appendix S5) additionally clustering two-way, by study and by underlying database. At the extremes of no winsorization and 5%, the bias-corrected short-run estimate is -0.1696 and -0.1683, the long-run estimate -0.4639 and -0.3572, so the 1% choice drives none of the headline conclusions.

The identification ladder. The key variable is the *price-identification strategy* of each estimate, coded on a six-tier ladder: (1) randomized experiments and randomized encouragement designs; (2) mandated natural experiments with exogenous price variation; (3) difference-in-differences designs; (4) instrumented specifications (IV, 2SLS/3SLS, GMM); (5) panel fixed effects and structural demand systems without instrumentation; (6) naive time-series or cross-section regressions with no attempt to address price endogeneity. Coding rules and study-level examples appear in Supplementary appendix S3. Because tiers 1–3 are thin in this literature (136, 10, and 26 estimates respectively) we group them as *design-based* throughout the main text; 878 estimates whose strategy cannot be classified from the primary study are excluded from

Figure 1: Nine decades of estimates, and no visible drift



Notes: Reported own-price elasticities against the mid-year of the underlying data, colored by identification tier; the 3,843 tier-classified estimates with usable standard errors are shown, the vertical axis truncated at $[-2.5, 1]$ so that 44 outlying estimates fall outside the frame; these outliers are omitted from the figure only and kept in every analysis (unclassified estimates are likewise omitted from the plot but not from non-ladder analyses). Two features organize the paper: the raw record shows no visible drift over nine decades, and the design-based literature (dark blue) arrives late and thin (8 studies contribute short-run design-based estimates), so composition must be held fixed before any trend is read as behavior.

ladder analyses and retained elsewhere. In the coded corpus, sixteen studies report estimates on more than one tier.

Adjustment horizon and the two clocks. Each estimate is classified as short-run (within roughly a year), intermediate-run, or long-run, using the primary study's own labels where given and the model structure otherwise (static single-equation estimates on levels with cointegration pre-tests are long-run; dynamic models contribute both a short- and a long-run estimate). We treat the horizon as an object of interest in its own right: Section 5 estimates the elasticity as a function of it.

Finally, every estimate carries two dates: the mid-year of the data used (the *data clock*; 23 estimates with no reported data years are assigned publication year minus three) and the publication year (the *publication clock*). Only changes in consumer behavior can shift the first; the second also shifts with publication practices, methods fashion, and selection pressure. Most previous surveys used one or the other; Section 6 uses both at once.

Table 1: Well-identified estimates have smaller short-run response

Tier	N	Studies	Unadjusted	Adjusted	95% CI
Design-based (RCT/nat. exp./DID)	81	8	-0.091	0.022	[-0.21, 0.25]
Instrumented (IV/GMM)	549	52	-0.304	-0.195	[-0.35, -0.04]
Panel FE / structural	655	98	-0.209	-0.126	[-0.28, 0.03]
Naive	316	70	-0.365	-0.238	[-0.37, -0.11]

Notes: The table presents the short-run price elasticity by identification tier. Headline sample (Marshallian-equivalent, usable SE), short run; 1,601 estimates from 220 studies enter the adjusted regression (tier-classified subset). Unadjusted is the raw mean within the tier. Adjusted is the conditional mean from one meta-regression of the elasticity on its standard error, tier indicators (naive omitted), and study characteristics, evaluated at SE = 0 and characteristics at sample means; 95% study-clustered intervals. Relative to naive, the design-based shift is +0.260 ($p = 0.006$), the fixed-effects shift +0.112 ($p = 0.003$), the IV shift +0.043 ($p = 0.32$). RCT, randomized controlled trial; nat. exp., natural experiment; DID, difference-in-differences; IV, instrumental variables; GMM, generalized method of moments; FE, fixed effects.

3 The identification ladder

Does the elasticity survive scrutiny of its identification? Table 1 lines the short-run literature up along the ladder. The unadjusted column reports the raw mean within each tier; the adjusted column reports the conditional mean from a single precision-controlling meta-regression: elasticity on its standard error (absorbing small-study effects), tier indicators, and a common set of study characteristics (sector, price measurement, data structure, region, demand controls, functional form), evaluated at a zero standard error with other characteristics at sample means, with study-clustered 95% intervals. This is a descriptive cross-study association: which design a study uses is not randomly assigned. We let the within-study evidence and the design-based benchmark carry the causal weight; the design-based benchmark itself rests on the cross-study comparison.

The pattern is a gradient with one exception. Raw means fall in magnitude from -0.37 (naive) to -0.09 (design-based); adjusted, the design-based tier sits at $+0.02$ with an interval of $[-0.21, 0.25]$, and is significantly less elastic than the naive baseline. That interval is wide: we read the design-based cell as consistent with a small elasticity while unable to exclude a moderate one. In the menu (Table 5), -0.09 is the raw and PET value and $+0.02$ the composition-adjusted prediction. The exception is instrumenting: the IV tier is the only identification upgrade that stays as elastic as the naive literature after adjustment (-0.19 against -0.24 naive), statistically indistinguishable from naive and overshooting the design-based benchmark. Within the sixteen studies that report estimates on more than one tier, the same picture appears in miniature: the within-study contrast between the cleaner and the lower tier averages −0.08 (cleaner estimates *more* elastic, driven by IV-versus-naive pairs), but it is small relative to its dispersion ($t = -0.79$) and positive in half the studies. The within-study evidence is underpowered, so we report it for completeness without leaning on it (Figure S4 in Supplementary appendix S4 plots every contrast). None of these within-study contrasts involves a design-based estimate, so the within-study evidence bears on the IV and fixed-effects ranking rather than on the design-based benchmark, and the argument depends on the cross-study ladder and the funnel evidence of the next section.

Two readings of the IV overshoot are possible, and the data cannot separate them here: instruments may correct attenuation from classical average-price measurement error (Alberini & Filippini, 2011), legitimately raising magnitudes; or weak instruments and specification search may inflate them. Ito (2014) shows the average-price coefficient is itself the behaviorally relevant object rather than an attenuated one, which cuts against the attenuation-correction reading. What the ladder does show is the direction that matters for the flexibility debate: *no identification upgrade makes demand look more price-responsive than the naive literature already claims*: the best designs point toward less responsiveness.

Price measurement provides a second check on the ladder. Average-price estimates carry classical division bias and inflated standard errors; marginal-price estimates are cleaner but scarcer. Bias-corrected short-run elasticities are −0.15 (marginal price) and −0.17 (average price): the correction for measurement regime moves the consensus by two hundredths, far less than the flexibility premise requires.

4 Selective reporting along the ladder

A meta-analysis that ignores selective reporting conflates the literature's publication preferences with the underlying parameters (Stanley *et al.*, 2008; Ioannidis *et al.*, 2017). The concern is concrete here: intuitive negative elasticities and statistically significant elasticities are easier to publish, which leaves a correlation between estimates and standard errors that funnel-based estimators detect and net out.

Table 2 reports the full battery of tests. First, selection is real and it inflates: the short-run funnel-asymmetry test rejects strongly (FAT = −0.85, $p < 0.001$), and every corrector

Table 2: Publication bias inflates the elasticities across tiers and horizons

Sample	N	Studies	Mean	PW	PET	PEESE	WAAP
Long-run elasticities (full sample)	723	151	−0.583	−0.378	−0.377	−0.498	−0.376
Intermediate elasticities (full sample)	954	108	−0.440	−0.149	−0.331	−0.416	−0.148
Short-run elasticities (full sample)	1,647	226	−0.267	−0.114	−0.163	−0.231	−0.111
Short-run, design-based	81	8	−0.091	−0.078	−0.095	−0.114	−0.080
Short-run, instrumented (IV)	549	52	−0.304	−0.104	−0.203	−0.276	−0.102
Short-run, panel FE / structural	655	98	−0.209	−0.150	−0.088	−0.164	−0.151
Short-run, naive	316	70	−0.365	−0.108	−0.245	−0.318	−0.106
Short-run, marginal price	401	48	−0.301	−0.054	−0.148	−0.230	−0.049
Short-run, average price	1,030	166	−0.264	−0.107	−0.168	−0.236	−0.105
Short-run, preferred only	426	191	−0.302	−0.159	−0.182	−0.271	−0.158

Notes: The table presents the publication-bias battery by horizon and tier. Headline sample; the full-sample rows pool all estimates at each horizon, below which the short-run sample is split by tier and by price regime. Mean is the raw mean; PW the precision-weighted mean; PET and PEESE the funnel-asymmetry-corrected estimates (Stanley, 2005; Stanley *et al.*, 2008); WAAP the weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Study-clustered inference. FAT p-values for the asymmetry test: short-run all < 0.001; design-based 0.97; marginal-price subsample 0.001. "Preferred only" uses the estimates flagged as preferred by the original studies (426 estimates from 191 studies). The sample-size-weighted corrector on the short-run full sample (1,647 estimates with a usable weight) gives −0.04 (se 0.032), the range's low end.

pulls the raw mean of −0.27 toward zero, to −0.16 (PET), −0.23 (PEESE), −0.11 (WAAP), −0.10 (top decile by precision), −0.04 (se 0.032; 1,647 estimates; sample-size weighted); a fuller battery in Supplementary appendix S5, adding the MAIVE estimator of Irsova *et al.* (2025), p-uniform* (van Aert & van Assen, 2026), and a selection model (Andrews & Kasy, 2019), stays in the same range. We report the range without adjudicating among correction philosophies; the corrected short-run consensus lies between −0.04 and −0.23, and nowhere near the raw mean. Second, it is the *observational* literature that shows selection bias: the design-based sample shows no funnel asymmetry (FAT $p = 0.97$), though with 81 estimates from 8 studies the test is under-powered, so selection can be neither detected nor excluded there; we rest the design-based benchmark on design credibility rather than on this null FAT. The slope itself is small ($\gamma = 0.029$) and stays insignificant under few-cluster-robust inference (wild cluster bootstrap $p = 0.918$ Rademacher, $p = 0.941$ Webb; CR3-Satterthwaite $p = 0.991$), so the null is not an artifact of the asymptotic approximation with only 8 clusters. The cell's bias-corrected mean is, in the same way, indistinguishable from zero under the wild-cluster bootstrap (-0.09, $p = 0.07$, 95% CI $[-0.23, 0.04]$).[1] Leave-one-study-out sharpens this caveat rather than

[1]The other thin-cluster cell the analysis leans on, the most recent long-run data decade (Figure 3, 2010s onward, 9 studies), is robust in the opposite direction: its bias-corrected mean, if anything more elastic than the −0.29 to −0.47 era band, stays firmly negative under the same bootstrap ($p = 0.002$, 95% CI $[-0.72, -0.38]$), so the strength of the recent long-run response is not a few-cluster artifact.

resolving it: dropping any one of the 8 studies moves the FAT slope as low as −1.96 and the p-value as high as 0.99, so the design-based null is a property of the full 8-study cell, not a result that survives study-by-study scrutiny. Third, the asymmetry is not an artifact of constructed standard errors or average-price measurement error: it survives, essentially undiminished, in the marginal-price subsample (FAT $= -0.97$, $p = 0.001$), which is the diagnostic that separates genuine selection from errors-in-variables mechanics. A caliper test points the same way from a different angle: there is no excess mass of estimates just above the conventional $|t| = 1.96$ significance threshold, even in the observational subsample (share above within a ± 0.2 window $= 0.51$, $p = 0.87$; full headline sample 0.53, $p = 0.39$), so the selective reporting this literature displays acts on the magnitude of estimates (a small-study effect) rather than on crossing the significance threshold. The two tests are complementary rather than competing: a literature filtered on the sign and size of estimates, not on their p-values, is precisely one that shows funnel asymmetry without threshold bunching.

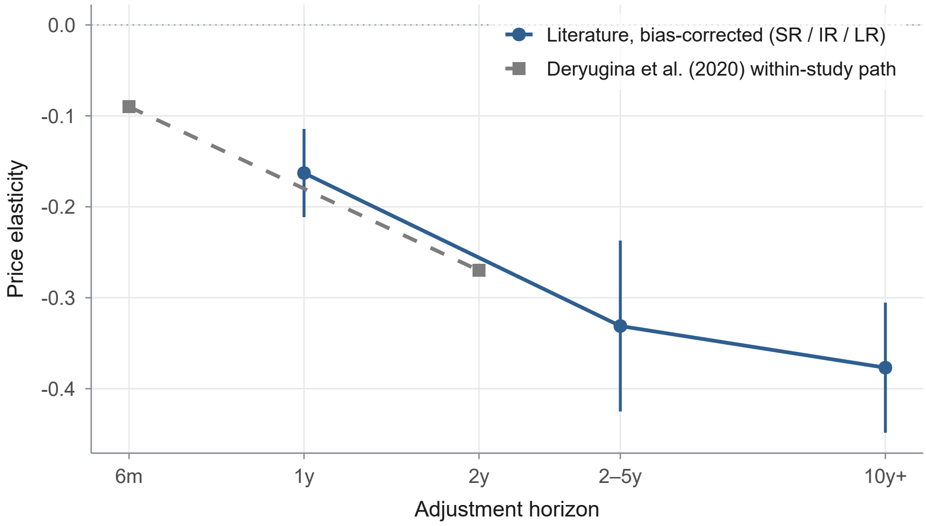
Because the correctors themselves rely on the standard errors that the selection story calls into question, we triangulate with weights that do not use them: weighting by sample size yields a corrected short-run elasticity of −0.04 (se 0.032; 1,647 estimates), statistically indistinguishable from zero and the least elastic value in the battery. Restricting to the estimates the original studies flag as preferred (426 estimates from 191 studies) leaves the picture unchanged (PET −0.18), as does the long run on the same restriction (251 estimates from 137 studies; PET −0.36), and the calendar-time trend on the preferred-only short-run sample stays indistinguishable from zero (−0.056 per decade, $p = 0.084$), if anything more negative than the pooled headline. All of these checks converge: *the bias-corrected short-run price elasticity of electricity demand is small*, and every standard correction moves it closer to zero.

5 Responsiveness grows with adjustment time

If short-run responsiveness is small, the flexibility case shifts to the long run: given time, capital adjusts. The record supports this mechanism and quantifies it.

Figure 2 plots the bias-corrected elasticity by horizon: −0.16 in the short run, −0.33 in the intermediate run, −0.38 in the long run: the response roughly doubles as adjustment time accumulates and then plateaus. This is not an artifact of which studies happen to populate

Figure 2: The bias-corrected elasticity grows with the adjustment horizon, tracking the quasi-experimental benchmark



Notes: Bias-corrected (PET) elasticity by horizon on the headline sample, with study-clustered 95% intervals: short run −0.163 (se 0.025; 1,647 estimates, 226 studies), intermediate run −0.331 (se 0.048; 954/108), long run −0.377 (se 0.037; 723/151). Dashed: the within-study adjustment path of Deryugina *et al.* (2020), −0.09 at six months to −0.27 at two years. Horizon placement of the literature bins is notational (SR ≈ within a year; IR 2–5 years; LR beyond).

which horizon bin: among the 100 studies that report both a short- and a long-run estimate, the same-study contrast goes from −0.238 to −0.593, a within-study ratio of means of 2.492 (median study-level ratio 2.571), with 86.0% of studies showing a more elastic long run and a paired t of −8.68. A within-study fixed-effects regression on the 1,241 estimates from the 112 studies reporting more than one horizon, with study fixed effects absorbing every cross-study difference in composition, puts the same steepening at −0.12 (se 0.065) from short to intermediate run and −0.22 (se 0.028) from short to long run: the pooled gradient is not a compositional artifact of which studies populate which bin.

Two studies help interpret the adjustment profile. Within a single natural experiment, Deryugina *et al.* (2020) track the same set of over 250 municipally-aggregated communities in a difference-in-differences design and find the elasticity tripling from −0.09 at six months to −0.27 after two years: the literature-wide gradient is directionally consistent with what this dynamic design delivers, a rare point of agreement between this literature and design-based work. Because this study also supplies the short-run design-based cell, it grounds the horizon path but is not independent corroboration of both findings. And the plateau has a mechanism: Costa & Kahn (2011) show that electricity consumption is embodied in the housing stock

Table 3: The horizon gradient is stable across eras of data

Horizon	≤1975	1976–90	1991–05	≥2006
Short run	−0.179	−0.090	−0.158	−0.223
	(0.038)	(0.041)	(0.034)	(0.043)
Intermediate run	−0.418	−0.256	−0.258	−0.241
	(0.080)	(0.081)	(0.060)	(0.112)
Long run	–	−0.429	−0.287	−0.472
		(0.057)	(0.046)	(0.055)

Notes: Bias-corrected (PET) elasticity within horizon × era cells, headline sample, study-clustered standard errors in parentheses. The long-run cell before 1976 has 51 estimates from 12 studies but is imprecisely estimated (a wide interval) and is suppressed. The intermediate run varies non-monotonically across eras; the short- and long-run values in the most recent era are not statistically distinguishable from the earliest era *shown* (for the long run this baseline is 1976–90, because the ≤1975 long-run cell is suppressed as imprecise), and no horizon exhibits a monotone rise toward the present. These era comparisons are imprecise and should be read as a limit on power rather than as evidence of exact stability.

at construction (an implied elasticity of about −0.22 at construction time), so the long-run response works through the turnover of buildings and appliances and stops at the substitution possibilities the stock allows. Descriptive dynamics in the literature point the same way: the median adjustment speed among dynamic models implies half of the long-run response within roughly a year and a half and ninety percent within five and a half years (Dahl, 2011). Figure S5 in Supplementary appendix S4 draws this adjustment path as a single continuous curve, pinned to the corpus's own short- and long-run values and consistent with these speeds.

The policy-relevant corollary follows directly: *the long run is already in the data.* A planner invoking "the long run" is invoking a central estimate of −0.38, not the unit-elastic −1 that fully flexible demand would imply; we note that this long-run value comes from the observational correctors (the design-based long-run cell is a single study, too thin to report), and that the within-study long-run projection of Deryugina *et al.* (2020) carries a 95% interval reaching −1, so −0.38 is a central estimate rather than a design-audited bound. The gradient itself is stable across eras: within every data-vintage era from the 1970s to the 2010s, the long-run corrected elasticity sits between −0.29 and −0.47 (Table 3). In the pooled record, adjustment time roughly doubles responsiveness (era-by-era the gradient varies with the short-run dips of the 1980s but shows no tendency to steepen toward the present).

Table 4: Two clocks, and neither shows a trend

	Data clock	Publication clock
Short run		
Data clock only	−0.018 (0.013)	–
Two clocks	−0.033 (0.031)	0.019 (0.033)
Two clocks + composition	0.017 (0.027)	−0.030 (0.028)
+ country fixed effects	0.044 (0.033)	−0.045 (0.033)
Intermediate run		
Data clock only	0.012 (0.028)	–
Two clocks	0.000 (0.077)	0.012 (0.065)
Two clocks + composition	0.043 (0.064)	0.030 (0.057)
+ country fixed effects	0.053 (0.065)	−0.003 (0.060)
Long run		
Data clock only	0.011 (0.036)	–
Two clocks	−0.092 (0.048)	0.135 (0.047)
Two clocks + composition	−0.019 (0.053)	0.073 (0.050)
+ country fixed effects	0.076 (0.056)	0.007 (0.058)

Notes: Each cell is a coefficient per decade from a study-clustered meta-regression that always controls for the standard error. The *Data clock* column is the trend in the elasticity per decade of the underlying *data* (do consumers behave differently over time?); the *Publication clock* column is the trend per decade of *publication* year (does the literature report differently over time?). The four rows are nested specifications: *Data clock only* includes just the data clock; *Two clocks* adds the publication clock, so the two are separated; *Two clocks + composition* further adds the identification tier and study characteristics of Table 1, that is, the changing *composition* of the literature, leaving any residual data-vintage trend. The final row within each block, + *country fixed effects*, adds a full set of country dummies to that specification, so the data-vintage trend is identified only from within-country variation and cannot reflect the changing country mix of the literature; it stays insignificant at every horizon and, if anything, turns more positive (short run +0.044, $p = 0.19$; long run +0.076, $p = 0.17$), further from the negative slope a rising-flexibility world would require. Unlike Table 1, which drops the estimates whose strategy could not be classified (leaving 1,601 in the short run), the composition-adjusted rows retain them in the naive reference category, so every row here is estimated on the full sample. Short-run sample: 1,647 estimates, 226 studies, data span 63 years; intermediate run: 954 estimates, 108 studies; long-run: 723 estimates, 151 studies, 54 years. A negative coefficient means the literature moves toward more elastic (more negative) values. No linear data-clock coefficient is significant at 5% in any specification. Quadratic specifications yield a significant squared term in the long run ($p < 0.001$) but not the short run (squared $p = 0.06$); we therefore label the long-run trend inconclusive rather than flat. The short-run quadratic does carry a significant linear term (−0.030 per decade, $p = 0.03$), but every linear short-run specification above is insignificant, consistent with the non-monotone mid-decade swings of Figure 3 rather than a trend.

6 The calendar-time test

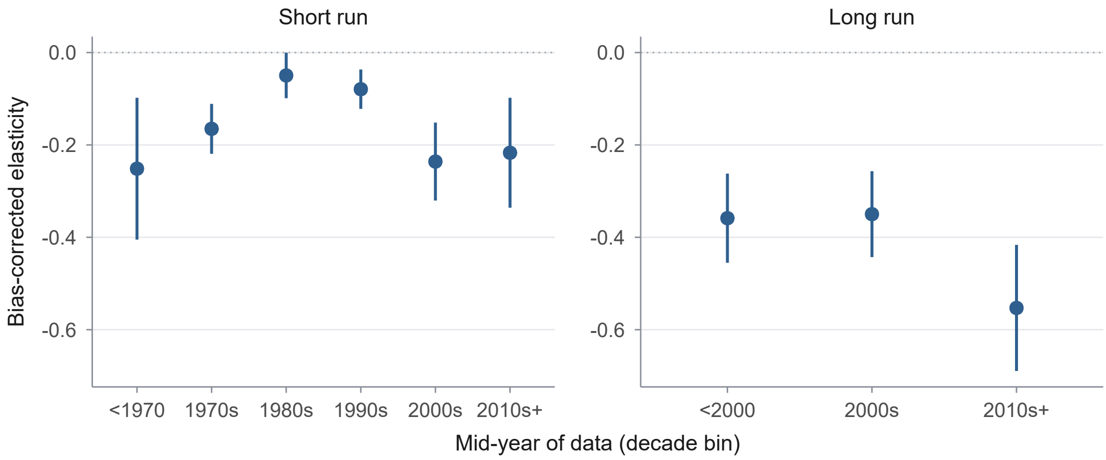
A small measured elasticity could still be reconciled with the flexibility premise in three ways. The true response could be larger, hidden by weak identification. Responsiveness could have grown over time, with selective reporting concealing the rise. Or the extra response could still be to come, arriving only in the long run. Sections 3–5 closed off all three. What remains is the premise's core empirical claim, that responsiveness has been rising as technology diffuses, and it is directly testable on the data clock.

Table 4 reports the trend under progressively stricter specifications; Figure 3 shows the decade-by-decade profile. The answer is the same throughout. In the raw record, the short-run trend is -0.010 per decade of data ($p = 0.43$). Controlling for precision, -0.018 ($p = 0.17$). Separating the two clocks leaves neither significant: the data clock is -0.033 ($p = 0.27$) and the publication clock $+0.019$ ($p = 0.57$); jointly, the two calendar coefficients are indistinguishable from zero as well (Wald $\chi^2_2 = 2.18$, $p = 0.34$), so the modest apparent drift that earlier surveys read as "elasticities changing over time" (Espey & Espey, 2004; Zabaloy & Viego, 2022) splits into two pieces, each indistinguishable from zero, rather than a behavioral trend in consumers. Holding identification and composition fixed, the data-clock point estimate turns positive: $+0.017$ per decade ($p = 0.53$),[2] if anything, a drift toward *less* price-responsiveness as the credibility revolution is netted out. The sign convention matters here: a rising-flexibility world requires this coefficient to be robustly *negative*. The pooled coefficient could still mask a rise concentrated in the settings the premise is built on, so we interact the data clock with sector and region and re-estimate it within each: none is robustly negative. The residential interaction is not distinguishable from the pooled trend (-0.021 per decade, $p = 0.30$), and the composition-adjusted data-clock slope is negative but insignificant in Europe (-0.067, $p = 0.18$), in post-2000 data (-0.037, $p = 0.60$), and in the residential-and-post-2000 intersection the premise is specifically about (-0.079, $p = 0.35$); the one subsample where the slope is significant, the United States ($+0.099$, $p = 0.002$), carries the wrong sign for a rising-flexibility world. We detect no rise in the settings that matter most to the premise.

The bound. Statistical insignificance alone would say little; what matters is the width of the interval. Take the canonical flexibility scenario: short-run responsiveness roughly doubles over three decades, from -0.16 to -0.33, an increase of about 0.16 in magnitude. Specification by specification, the edge of the 95% interval most favorable to that scenario allows a cumulative three-decade increase of 0.10 (raw record, 64% of the required movement), 0.13 (precision-controlled bivariate, 80%), and 0.11 (two clocks plus composition, 66%). The one exception is instructive: splitting the two clocks *without* composition controls inflates the data-clock standard error enough (the two clocks are collinear) that its interval alone could accommodate a doubling, which is why the composition-adjusted specification is the informative one. In

[2] Entering the controls one block at a time (data clock alone, then the publication clock, then the identification tier, then study composition) confirms the point estimate: $+0.017$ ($p = 0.53$) at the full, composition-adjusted specification, identical to the headline.

Figure 3: Bias-corrected elasticity by decade of data



Notes: PET-corrected elasticity within decade bins of the data mid-year, headline sample, 95% study-clustered intervals. Short run: −0.251 (<1970), −0.165 (1970s), −0.050 (1980s), −0.079 (1990s), −0.236 (2000s), −0.217 (2010s+). The 2010s+ value is statistically indistinguishable from the 1970s value; the series swings across the middle decades (the 1980s and 2000s intervals do not overlap). Long run: −0.36, −0.35, −0.55 across coarse bins, the last on 9 studies.

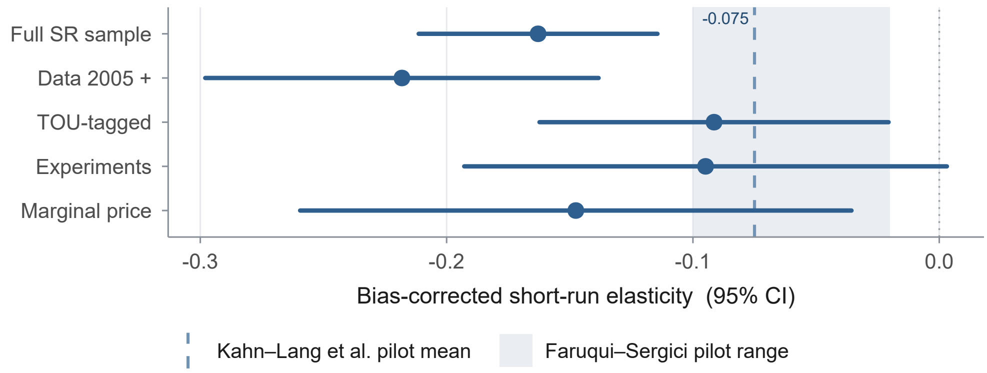
every specification precise enough to bound change, the record does more than fail to show the assumed rise: it bounds any concealed rise below what the scenario needs.

The formal version of that bound is an equivalence test. The doubling scenario requires responsiveness to rise by 0.05428 per decade, a full doubling spread evenly across three decades. The test asks, one-sided, whether the data are still consistent with a rise that large. In all three specifications they are not: it rejects a doubling-sized rise at the 5% level ($p = 0.0002$ raw, $p = 0.0026$ precision-controlled, $p = 0.004$ two clocks plus composition). So the exclusion rests on a test, not on eyeballing where an interval ends. How large a rise could each specification have caught in the first place? At 80% power, the two precise specifications would detect a rise of just 65–67% of a doubling, so even a partial rise would not have slipped past them. The composition-adjusted specification is noisier: measured against zero, it could only detect a rise larger than a full doubling (138%). It still rejects the doubling anyway, because the equivalence test measures the estimate against the rise margin, not against zero, and the estimate is positive, pointing the opposite way and so sitting comfortably clear of that margin. The long-run record, being noisier (a per-decade interval of ±0.10 around zero and a significant quadratic), cannot support a symmetric claim, and we label it inconclusive at current power rather than flat.

The new-regime objection. The record's data end in 2024, and the premise is about the future: perhaps smart meters, automation, and time-varying rates change the regime after the sample. We make no claim about that post-sample future. But the settings the premise is built on already appear in the record. Where they do, demand is no more price-responsive than elsewhere. The objection can be addressed inside the data, because the record already contains the tariff instrument central to the premise, time-of-use pricing. The record covers the pricing instrument rather than the full enabling-technology regime (mass smart metering, automation, and storage) that the premise projects for the post-sample future, and it samples that regime thinly. We therefore read the result below as suggestive of the technology-rich corner rather than dispositive for it.

Figure 4 compares corrected short-run elasticities across cells: the full sample (-0.16), data from 2005 onward (-0.22), marginal-price studies (-0.15), time-of-use settings (-0.09), and design-based studies (-0.09; the field experiments of Jessoe & Rapson, 2014 and Ito *et al.*, 2018, and the natural experiment of Deryugina *et al.*, 2020); the broader pilot literature concurs, with the -0.02 to -0.10 range of Faruqui & Sergici (2010) and the -0.075 average that Kahn-Lang *et al.* (2025) extract from post-2000 U.S. pilots. That same pilot literature reports substitution elasticities of 0.07 to 0.40 and peak own-price values as large as -0.79. Those describe intertemporal load-shifting, which is outside the own-price level object we measure. In the meta-regression, the time-of-use indicator enters at $+0.14$ ($p = 0.007$): conditional on precision, estimates from the most technologically enabled settings are closer to zero than the literature at large. The model-averaged counterpart is smaller and less firmly selected ($+0.07$, PIP $= 0.73$; Table S14), so we read the time-of-use signal as directionally consistent but less firmly identified than the design-based tier. Nothing in the technology-rich corner of the record hints at a rise in the own-price *level* elasticity we measure. The metric is narrow by construction: an own-price level elasticity does not capture intraday load-shifting or intertemporal substitution, which net out in total consumption, so these cells bound the level response but are silent on engineered demand-response flexibility. What does move pilot responses is hardware and program design (enabling technology adds over twenty percentage points to peak reductions in Kahn-Lang *et al.* (2025)), which supports our central conclusion: flexibility comes from engineering, and prices alone do not deliver it.

Figure 4: The technology-rich cells are the least elastic



Notes: PET-corrected short-run elasticity with 95% study-clustered intervals, by cell of the headline sample (n: full 1,647; post-2005 499; TOU 210; experiments 81; marginal price 401). Shaded band: the own-price elasticity range from dynamic-pricing pilots surveyed by Faruqui & Sergici (2010); dashed line: the −0.075 pilot average of Kahn-Lang *et al.* (2025).

Table 5: A menu of corrected consensus elasticities for planning and modeling

Object	Preferred	Range across correctors
Short run, full sample	−0.16	−0.04 to −0.23
Short run, design-based	−0.09	−0.08 to −0.11
Short run, time-of-use settings	−0.09	–
Intermediate run	−0.33	−0.15 to −0.42
Long run	−0.38	−0.38 to −0.50
Trend allowance, per decade	0.00	95% interval (−0.04, +0.07) adj. (−0.04, +0.01) bivariate

Notes: Bias-corrected values on the headline sample (ranges across PET, PEESE, WAAP, precision- and sample-size weighting where computed). The trend allowance row reports the 95% interval on the short-run data clock in the composition-adjusted specification and, for comparison, the precision-controlled bivariate one; the unrestricted two-clock split is too collinear to be informative (data and publication clocks correlate at $r = 0.915$, VIF $= 6.12$, inflating the data-clock standard error 2.35×; see Section 6).

Aigner's conjecture (Aigner, 1985) can now be answered. Households replaced their appliances several times over, metering and displays arrived at scale, and automated response began its rollout. Four further decades of estimates accumulated, and the predicted growth in responsiveness never materialized. The conservation effect the early experiments measured turned out to be, as Aigner himself suspected it might, the full extent of the response.

7 What should planners and models use

Energy-system models rarely document their demand elasticities; where visible, household own-price values of −0.2 to −0.6 in the long run are common (Huntington *et al.*, 2019). Table 5 states what the audited record supports. The menu implies three planning rules. First, *match* *the elasticity to the horizon*: −0.16 for operations within a year, −0.38 for capital-turnover horizons. The upper end of the modeled range (beyond −0.5) has little support in the corrected record. These menu values are residential-dominated; the long-run response is roughly half as large in industrial and commercial settings (Table S2), so a whole-system planner should apply a demand-weighted blend smaller in magnitude than −0.38. Second, *any assumed growth in responsiveness is a technology assumption rather than a demand assumption*: the historical record caps organic growth at roughly 0.04 per decade in magnitude in every specification precise enough to bound change, so a scenario in which the effective elasticity doubles must specify the hardware, contracts, and automation that deliver the difference, and should price them, because the same pilot literature that finds small elasticities finds large effects of enabling technology. Third, *treat headline literature means as upper bounds*: the raw mean of −0.27 reflects selective reporting; every correction philosophy and every identification upgrade moves the number toward zero. For guidance below the aggregate, Table S18 reports a best-practice elasticity for each country with at least five headline estimates, the value an ideally-designed study would recover there; the cross-country spread is modest, and no country's profile supports a large or rising response.

8 Conclusion

We asked whether the price responsiveness of electricity demand has risen, as four decades of energy planning has assumed it would. Across 4,720 estimates from 462 studies spanning nine decades of data, the answer is no. The corrected short-run elasticity is small (−0.16 corrected, −0.09 in design-based studies, indistinguishable from zero once composition is held fixed), and nearly every step up the identification ladder, every publication-bias correction, and every technology-rich subsample moves it closer to zero; the one exception is the post-2005 data cell (−0.22), which is more elastic than the full sample yet still sits inside the corrected range and shows no upward trend. Responsiveness does grow with adjustment time, doubling toward −0.38 as the durable stock turns over, exactly as the quasi-experimental adjustment path and the embodied-capital mechanism predict; but that gradient has been in the data since the 1970s and has not steepened. And on the clock that matters (the vintage of the data, separated from the vintage of the publications), the elasticity shows no trend: in the specifications precise enough to bound it, the record caps any concealed three-decade rise at two-thirds to four-fifths of what a doubling scenario assumes, and the point estimate leans the other way.

We do not claim that responsiveness cannot rise. Automation that responds to prices without human attention, storage that arbitrages tariffs, and contracts that delegate curtailment are all technologies that create price responsiveness, and the pilot record shows they work. Our claim is about the assumption that does the planning work today: demand has never yet become more price-elastic merely because pricing became more sophisticated, and a half-century of evidence limits how much such a change could have gone undetected. Prices alone have never made electricity demand flexible, and we find no evidence that this is changing.

Three limitations bound these conclusions. First, holding identification and composition fixed moves the short-run data clock from -0.033 to $+0.017$. Because design quality is itself correlated with data vintage, the composition controls could in principle absorb a genuine behavioral trend, though the raw record independently shows no trend, which mitigates this concern. Second, the corrected values are read at a zero standard error, an out-of-support extrapolation of the funnel. Third, the design-based benchmark rests on eight short-run studies.

Use of artificial intelligence

Claude Opus 4.8 by Anthropic and GPT 5.6 Sol by OpenAI assisted in cross-checking the findings and editing the text, following the guidance of Cook *et al.* (2026a) and Cook *et al.* (2026b) on the use of AI in meta-analysis. All results were produced by running the analysis code on the dataset and can be reproduced from the public replication package. The authors are responsible for all of the paper's content.

References

van Aert, R. C. & M. A. van Assen (2026): "Correcting for publication bias in a meta-analysis with the p-uniform* method." *Psychonomic Bulletin & Review* **33(3)**: p. 102.

Aigner, D. (1985): "The residential electricity time-of-use pricing experiments: what have we learned?" In "Social experimentation," pp. 11–54. University of Chicago Press.

Alberini, A. & M. Filippini (2011): "Response of residential electricity demand to price: The effect of measurement error." *Energy Economics* **33(5)**: pp. 889–895.

Andrews, I. & M. Kasy (2019): "Identification of and correction for publication bias." *American Economic Review* **109(8)**: pp. 2766–2794.

Brodeur, A., N. Cook, & A. Heyes (2020): "Methods matter: p-hacking and publication bias in causal analysis in economics." *American Economic Review* **110(11)**: pp. 3634–3660.

Christensen, G. & E. Miguel (2018): "Transparency, reproducibility, and the credibility of economics research." *Journal of Economic Literature* **56(3)**: pp. 920–980.

Cook, N., F. Bartos, P. R. D. Bom, S. Gechert, K. Kantova, J. Geyer-Klingeberg, T. Havranek, Z. Irsova, M. Luskova, M. Opatrny, F. Prante, H. J. Rachinger, & T. D. Stanley (2026a): "Guidance for the use of ai in the meta-analysis of economics research." *Journal of Economic Surveys, forthcoming* .

Cook, N., F. Bartos, P. R. D. Bom, S. Gechert, K. Kantova, J. Geyer-Klingeberg, T. Havranek, Z. Irsova, M. Luskova, M. Opatrny, F. Prante, H. J. Rachinger, & T. D. Stanley (2026b): "Reporting Guidelines for Meta-Analysis in Economics—Updated for AI." *Journal of Economic Surveys, forthcoming* .

Costa, D. L. & M. E. Kahn (2011): "Electricity consumption and durable housing: Understanding cohort effects." *American Economic Review* **101(3)**: pp. 88–92.

Dahl, C. (2011): "A global survey of electricity demand elasticities." In "Institutions, Efficiency and Evolving Energy Technologies, 34th IAEE International Conference, June 19-23, 2011," International Association for Energy Economics.

Dahl, C. A. (1993): "A Survey of Energy Demand Elasticities in Support of the Development of the NEMS." *MPRA Paper 13962*, University Library of Munich, Germany.

DellaVigna, S. & E. Linos (2022): "Rcts to scale: Comprehensive evidence from two nudge units." *Econometrica* **90(1)**: pp. 81–116.

Deryugina, T., A. MacKay, & J. Reif (2020): "The long-run dynamics of electricity demand: Evidence from municipal aggregation." *American Economic Journal: Applied Economics* **12(1)**: pp. 86–114.

Espey, J. A. & M. Espey (2004): "Turning on the lights: A meta-analysis of residential electricity demand elasticities." *Journal of Agricultural and Applied Economics* **36(1)**: pp. 65–81.

Faruqui, A. & S. Sergici (2010): "Household response to dynamic pricing of electricity: A survey of 15 experiments." *Journal of Regulatory Economics* **38(2)**: pp. 193–225.

Fatima, Z. (2023): *Price & Income Elasticity of Residential Electricity Demand in Asia: A Meta-Analysis*. Master's thesis, Charles University, Prague.

Huntington, H. G., J. J. Barrios, & V. Arora (2019): "Review of key international demand elasticities for major industrializing economies." *Energy Policy* **133**: p. 110878.

Ioannidis, J. P. A., T. D. Stanley, & H. Doucouliagos (2017): "The Power of Bias in Economics Research." *The Economic Journal* **127**: pp. F236–F265.

Irsova, Z., P. R. Bom, T. Havranek, & H. Rachinger (2025): "Spurious precision in meta-analysis of observational research." *Nature Communications* **16**: p. 8454.

Ito, K. (2014): "Do consumers respond to marginal or average price? evidence from nonlinear electricity pricing." *American Economic Review* **104(2)**: pp. 537–563.

Ito, K., T. Ida, & M. Tanaka (2018): "Moral suasion and economic incentives: Field experimental evidence from energy demand." *American Economic Journal: Economic Policy* **10(1)**: pp. 240–267.

Jessoe, K. & D. Rapson (2014): "Knowledge is (less) power: Experimental evidence from residential energy use." *American Economic Review* **104(4)**: pp. 1417–1438.

Kahn-Lang, J., Y. Zhu, K. Palmer, & P. Cappers (2025): "Different prices for different slices: A meta-analysis of time-based electricity rates." *Working Paper 25-04*, Resources for the Future.

Labandeira, X., J. M. Labeaga, & X. Lopez-Otero (2017): "A meta-analysis on the price elasticity of energy demand." *Energy Policy* **102**: pp. 549–568.

Marques, V., D. Uhr, & J. Uhr (2024): "Residential electricity demand in Latin America: A meta-analysis." *International Journal of Energy Sector Management* **18(2)**: pp. 369–389.

Stanley, T. D. (2005): "Beyond Publication Bias." *Journal of Economic Surveys* **19(3)**: pp. 309–345.

Stanley, T. D., C. Doucouliagos, & S. B. Jarrell (2008): "Meta-regression analysis as the socio-economics of economics research." *The Journal of*

Socio-Economics **37(1)**: pp. 276–292.

Torriti, J. (2014): "A review of time use models of residential electricity demand." *Renewable and Sustainable Energy Reviews* **37**: pp. 265–272.

U.S. Bureau of Labor Statistics (2024): "Consumer expenditure surveys." U.S. Department of Labor, `https://www.bls.gov/cex/`. Electricity share of household expenditures; accessed July 2026.

Zabaloy, M. F. & V. Viego (2022): "Household electricity demand in Latin America and the caribbean: A meta-analysis of price elasticity." *Utilities Policy* **75**: p. 101334.

Zhu, X., L. Li, K. Zhou, X. Zhang, & S. Yang (2018): "A meta-analysis on the price elasticity and income elasticity of residential electricity demand." *Journal of Cleaner Production* **201**: pp. 169–177.